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Vadodara, Gujarat

NAAC GRADE A++

Finite Difference and Interpolation

Study Guide

Mr. Hardik Gupta
Department Of Applied Science & Humanities, PIT
Parul University

Unit-5 Finite Difference and Interpolation

Finite Differences:

Suppose that the function $y = f(x)$ is tabulated for the equally spaced values $x = x_0, x_0 + h, x_0 + 2h, ...x_0 + nh$ giving $y = y_0, y_1, y_2, ...y_n$. To determine the values of $f(x)$ or $f'(x)$ for some intermediate values of x, the following three types of differences are found useful:

(1) Forward Differences: The differences $y_1 - y_0, y_2 - y_1, ...y_n - y_{n-1}$ when denoted by $\Delta y_0, \Delta y_1, \Delta y_2, ..., \Delta y_{n-1}$ respectively are called the first forward differences, where Δ is the forward difference operator. Thus, the first forward differences are $\Delta y_r = y_{r+1} - y_r$

Similarly, the second forward differences are defined by $\Delta^2 y_r = \Delta y_{r+1} - \Delta y_r$ In general, $\Delta^p y_r = \Delta^{p-1} y_{r+1} - \Delta^{p-1} y_r$ defines the p th forward differences.

Forward difference table

Values of x	Values of y	1st diff.	2nd diff.	3rd diff.	4th diff.	5th diff.
x_0	y_0	$\Delta y_0\,(y_1 - y_0)$	$\Delta^2 y_0$			
$x_0 + h$	y_1	$\Delta y_1\,(y_2 - y_1)$	$\Delta^2 y_1$	$\Delta^3 y_0$	$\Delta^4 y_0$	
$x_0 + 2h$	y_2	$\Delta y_2\,(y_3 - y_2)$	$\Delta^2 y_2$	$\Delta^3 y_1$	$\Delta^4 y_1$	$\Delta^5 y_0$
$x_0 + 3h$	y_3	$\Delta y_3\,(y_4 - y_3)$	$\Delta^2 y_3$	$\Delta^3 y_2$		
$x_0 + 4h$	y_4	$\Delta y_4\,(y_5 - y_4)$				
$x_0 + 5h$	y_5					

(2) Backward Differences: The differences $y_1 - y_0, y_2 - y_1, ...y_n - y_{n-1}$ when denoted by $\nabla y_1, \nabla y_2, \nabla y_3, \ldots\ldots\ldots\ldots \nabla y_n$ respectively are called the first Backward differences, where ∇ is the Backward difference operator. Similarly, we define higher-order backward differences. Thus, we have $\nabla y_r = y_r - y_{r-1}$.

In general, $\nabla^p y_r = \nabla^{p-1} y_r - \nabla^{p-1} y_{r-1}$ defines the p th forward differences.

Backward difference table

Values of x	Values of y	1st diff.	2nd diff.	3rd diff.	4th diff.	5th diff.
x_0	y_0					
		∇y_1				
x_0+h	y_1		$\nabla^2 y_2$			
		∇y_2		$\nabla^3 y_3$		
x_0+2h	y_2		$\nabla^2 y_3$		$\nabla^4 y_4$	
		∇y_3		$\nabla^3 y_4$		$\nabla^5 y_5$
x_0+3h	y_3		$\nabla^2 y_4$		$\nabla^4 y_5$	
		∇y_4		$\nabla^3 y_5$		
x_0+4h	y_4		$\nabla^2 y_5$			
		∇y_5				
x_0+5h	y_5					

Relationship between operators:

(1) $\Delta\nabla = \Delta - \nabla = \nabla\Delta$

Solution:

$$\begin{aligned}\Delta\nabla f(x) &= \Delta[\nabla f(x)]\\ &= \Delta[f(x)-f(x-h)]\\ &= \Delta f(x)-\Delta f(x-h)\\ &= f(x+h)-f(x)-[f(x)-f(x-h)]\\ &= \Delta f(x)-\nabla f(x)\\ &= (\Delta-\nabla)f(x)\end{aligned}$$

(2) $\Delta+\nabla = \dfrac{\Delta}{\nabla}-\dfrac{\nabla}{\Delta}$

Solution:

$$\begin{aligned}\frac{\Delta}{\nabla}-\frac{\nabla}{\Delta} &= \frac{\Delta^2-\nabla^2}{\nabla\Delta}\\ &= \frac{(\Delta-\nabla)(\Delta+\nabla)}{\Delta-\nabla}\\ &= \Delta-\nabla\end{aligned}$$

(3) The shift operator E :

The operator E is defined as

$Ef(x) = f(x+h)$

$E^2 f(x) = E[Ef(x)] = f(x+2h)$

.

.

.

$E^n f(x) = f(x+nh)$

(4) $E = 1+\Delta$

Solution:

$$\begin{aligned}Ef(x) &= f(x+h)\\ &= f(x+h)-f(x)+f(x)\\ &= \Delta f(x)+f(x)\\ &= (\Delta+1)f(x)\end{aligned}$$

(5) $E\nabla = \Delta$

Solution:

$$\begin{aligned}E\nabla f(x) &= E(\nabla f(x))\\ &= E(f(x)-f(x-h))\\ &= Ef(x)-Ef(x-h)\\ &= f(x+h)-f(x)\\ &= \Delta f(x)\end{aligned}$$

(6) $(1+\Delta)(1-\nabla) = 1$

Solution:

$$\begin{aligned}(1+\Delta)(1-\nabla) &= 1-\nabla+\Delta-\Delta\nabla\\ &= 1-\nabla+\Delta-(\Delta-\nabla)\\ &= 1-\nabla+\Delta-\Delta+\nabla\\ &= 1\end{aligned}$$

(7) $\nabla = 1 - E^{-1}$ **Solution:** $1 - E^{-1} = 1 - (1+\Delta)^{-1}$ $= 1 - \frac{1}{1+\Delta}$ $= \frac{1+\Delta-1}{1+\Delta} = \frac{E\nabla}{E} = \nabla$	**(8)** $E = e^{hD}$ **Solution:** $Ef(x) = f(x+h)$ $= f(x) + hf'(x) + \frac{h^2}{2!} f''(x) + ...$ $= f(x) + hDf(x) + \frac{h^2}{2!} D^2 f(x) + ...$ $= \left[1 + hD + \frac{h^2 D^2}{2!} + ...\right] f(x)$ $= e^{hD} f(x)$

- **Relations between the various operators:**

In terms of	E	Δ	∇	δ	hD
E	------	$\Delta + 1$	$(1-\nabla)^{-1}$		
Δ		------	$(1-\nabla)^{-1} - 1$		
∇		$1 - (\Delta+1)^{-1}$	-----		
δ		$\Delta(\Delta+1)^{-1/2}$		-----	
μ		$\left(1 + \frac{\Delta}{2}\right)(\Delta+1)^{-1/2}$			
hD					--------

Example: Prove with the usual notations that $(E^{1/2} + E^{-1/2})(1+\Delta)^{1/2} = 2 + \Delta$

Solution: $(E^{1/2} + E^{-1/2})(1+\Delta)^{1/2} = (E^{1/2} + E^{-1/2})\ E^{1/2}$ = E +1 = 1+ Δ +1 = 2+ Δ

EXAMPLE: Write the forward difference table if

x:	10	20	30	40
y:	1.1	2.0	4.4	7.9

Solution:

X	Y	Δy	$\Delta^2 y$	$\Delta^3 y$
10	1.1			
		0.9		
	2.0		1.5	
20		2.4		-0.4
	4.4		1.1	
30		3.5		
	7.9			
40				

INTERPOLATION

Let the function $y = f(x)$ take the values $y_0, y_1, y_2, y_3, ..., y_n$ corresponding to the values $x_0, x_1, x_2, x_3, ..., x_n$ of x. The process of finding the value of y corresponding to any value of $x = x_i$ between x_0 and x_n is called interpolation. Thus, interpolation is a technique of finding the value of

a function for any intermediate value of the independent variable.s
The process of computing the value of the function outside the range of given values of the variable is called extrapolation. The study of interpolation is based on the concept of finite differences, which were discussed later.

Newton's forward interpolation formula:

Let the function $y = f(x)$ take the values $y_0, y_1, y_2, y_3, ..., y_n$ corresponding to the values $x_0, x_1, x_2, x_3, ..., x_n$ of x. Suppose it is required to evaluate $f(x)$ for $x = x_0 + rh$, where r is any real number.

$$y_r = f(x_0 + rh)$$
$$= E^r f(x_0)$$
$$= (1+\Delta)^r f(x_0)$$
$$= (1+\Delta)^r y_0$$
$$= \left[1 + r\Delta + \frac{r(r-1)}{2!}\Delta^2 + \frac{r(r-1)(r-2)}{3!}\Delta^3 + ...\right] y_0$$

[Using Binomial theorem]

$$= y_0 + r\Delta y_0 + \frac{r(r-1)}{2!}\Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!}\Delta^3 y_0 + ...$$

$$y_r = y_0 + r\Delta y_0 + \frac{r(r-1)}{2!}\Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!}\Delta^3 y_0 + ...$$

The above formula is known as **Newton's forward interpolation formula**.

Example 1: From the following table, estimate the number of students who obtained marks between 40 and 45.

Marks	30 - 40	40 - 50	50 - 60	60 - 70	70 - 80
Number of students	31	42	51	35	31

Solution:

Marks less than (x)	40	50	60	70	80
No. of students (f(x))	31	73	124	159	190

X	f(x)	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$	$\Delta^4 f(x)$
40	31				
		42			
50	73		9		
		51		-25	
60	124		-16		37
		35		12	
70	159		-4		
		31			
80	190				

We shall find y_{45} i.e. number of students with marks less than 45. Taking $x_0 = 40, x = 45$, we have

$p = \frac{x - x_0}{h} = \frac{5}{10} = 0.5 \qquad (\because h = 10)$

Newton's forward interpolation formula,

$$y_{45} = y_{40} + p\Delta y_{40} + \frac{p(p-1)}{2!}\Delta^2 y_{40} + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_{40} + \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_{40}$$

$$= 31 + 0.5 \text{ x } .42 + \frac{0.5(-0.5)}{2} \text{ x}.9 + \frac{0.5(-0.5)(-1.5)}{6} \text{ x } (-25) + \frac{0.5(-0.5)(-1.5)(-2.5)}{24} \text{ x}.37$$

$= 47.87$

The number of students with marks less than 45 is 47.87 i.e. 48. But the number of students with marks less than 40 is 31. Hence the number of students getting marks between 40 and 45 = 48-31=17.

Example 2: The table gives the distance in nautical miles of the visible horizon for the given heights in feet above the earth's surface:

$x = height$	100	150	200	250	300	350	400
$y = distance$	10.63	13.03	15.04	16.81	18.42	19.90	21.27

Find the values of y when

1. $\boldsymbol{x = 160 fts}$

Solution:

x	y	Δ	Δ^2	Δ^3	Δ^4
100	10.63	2.40	-0.39	0.15	-0.7
150	**13.03**	**2.01**	**-0.24**	**0.08**	**-0.05**
200	15.04	1.77	-0.16	0.03	-0.01
250	16.81	1.61	-0.13	**0.02**	
300	18.42	1.48	**-0.11**		
350	19.90	**1.37**			
400	**21.27**				

1. If we take $x_0 = 160$ $then$ $y_0 = 13.03, \Delta y_0 = 2.01, \Delta^2 y_0 = -0.24, \Delta^3 = 0.08, \Delta^4 y_0 = -0.05$

Since $x = 160$ and $h = 50$

$$\therefore p = \frac{x - x_0}{h} = \frac{10}{50} = 0.2$$

∴ Using Newton's forward interpolation formula, we get

$$y_{218} = y_0 + p\Delta y_0 + \frac{p(p-1)}{2!}\Delta^2 y_0 + \frac{p(p-1)(p-2)}{3!}\Delta^3 y_0 + \frac{p(p-1)(p-2)(p-3)}{4!}\Delta^4 y_0 + \cdots \ldots \ldots \ldots \ldots$$

$y_{160} = 13.03 + 0.402 + 0.192 + 0.0384 + 0.00168 = 13.46$ $bautical\ miles$

2. Since $x = 410$ is near the end of the table, we use Newton's backward interpolation formula.

$\therefore taking\ x_n = 400, p = \frac{x - x_n}{h} = \frac{10}{50} = 0.2$

Using the line backward difference,

Example 3: Find the cubic polynomial which takes the following values:

X	0	1	2	3
f(x)	1	2	1	10

Hence or otherwise evaluate f(4).

Solution: The difference table is

X	f(x)	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
0	1			
		1		
1	2		-2	
		-1		12
2	1		10	
		9		
3	10			

We take $x_0 = 0$ and $p = \frac{x-0}{h} = x \qquad (\because h = 1)$

$$f(x) = f(0) + p\Delta f(0) + \frac{p(p-1)}{2!}\Delta^2 f(0) + \frac{p(p-1)(p-2)}{3!}\Delta^3 f(0)$$

$$= 1 + x(1) + \frac{x(x-1)}{2}(-2) + \frac{x(x-1)(x-2)}{6}(12) = 2x^3 - 7x^2 + 6x + 1$$

which is the required polynomial.

To compute f(4) we take $x_n = 3, x = 4$ so that $p = \frac{x - x_n}{h} = 1 \qquad (\because h = 1)$

($\because$ Newton's backward interpolation formula)

$$f(4) = f(3) + p\Delta f(3) + \frac{p(p+1)}{2!}\Delta^2 f(3) + \frac{p(p+1)(p+2)}{3!}\Delta^3 f(3)$$

= 10+9+10+12 = 41.

Example 4: Using Newton's forward interpolation formula, find the value of $f(1.6)$.

x	1	1.4	1.8	2.2
$f(x)$	3.49	4.82	5.96	6.5

Solution: Let

$$x = 1.6,\ x_0 = 1\ , h = 0.4$$

$$r = \frac{x - x_0}{h} = \frac{1.6 - 1}{0.4} = 1.5$$

Difference Table:

x	$f(x)$	$\Delta f(x)$	$\Delta^2 f(x)$	$\Delta^3 f(x)$
1	3.49			
		1.33		
1.4	4.82		-0.19	
		1.14		-0.41
1.8	5.96		-0.6	
		0.54		
2.2	6.5			

By Newton's forward interpolation formula,

$$y_r = y_0 + r\Delta y_0 + \frac{r(r-1)}{2!}\Delta^2 y_0 + \frac{r(r-1)(r-2)}{3!}\Delta^3 y_0 + ...$$

$$y_{1.6} = f(1.6) = 3.49 + (1.5)(1.33) + \frac{(1.5)(1.5-1)}{2!}(-0.19) + \frac{(1.5)(1.5-1)(1.5-2)}{3!}(-0.41)$$

$$= 5.4393$$

<u>Newton's Backward interpolation formula:</u>

Let the function $y = f(x)$ take the values $y_0, y_1, y_2, y_3, ..., y_n$ corresponding to the values $x_0, x_1, x_2, x_3, ..., x_n$ of x. Suppose it is required to evaluate $f(x)$ for $x = x_0 + rh$, where r is any real number.

$$y_r = f(x_n + rh)$$

$$= E^r f(x_n)$$

$$= (E^{-1})^{-r} f(x_n)$$

$$= (1-\nabla)^{-r} f(x_n)$$

$$= \left[1 + r\nabla + \frac{r(r+1)}{2!}\nabla^2 + \frac{r(r+1)(r+2)}{3!}\nabla^3 + ...\right] y_n$$

[Using Binomial theorem]

$$= y_n + r\nabla y_n + \frac{r(r+1)}{2!}\nabla^2 y_n + \frac{r(r+1)(r+2)}{3!}\nabla^3 y_n + ...$$

$$y_r = y_n + r\nabla y_n + \frac{r(r+1)}{2!}\nabla^2 y_n + \frac{r(r+1)(r+2)}{3!}\nabla^3 y_n + ...$$

The above formula is known as **Newton's backward interpolation formula**.

Example 4: Using Newton's backward difference formula, construct an interpolating polynomial of degree 3 for the data: f(-0.75)=-0.0718125, f(-0.5)= -0.02475, f(-0.25)= 0.3349375, f(0)=1.10100. Hence, find f(-1/3).

Solution: The difference table is

X	y	Δy	$\Delta^2 y$	$\Delta^3 y$
-0.75	-0.0718 125			
		0.0470625		
-0.5	-0.02475		0.312625	
		0.3596875		0.09375
-0.25	0.3349375		0.400375	
		0.7660625		
0	1.10100			

We use Newton's backward difference formula

$$y(x) = y_3 + p\Delta y_3 + \frac{p(p+1)}{2!}\Delta^2 y_3 + \frac{p(p+1)(p+2)}{3!}\Delta^3 y_3$$

$$x_3 = 0 \text{ and } p = \frac{x-0}{h} = \frac{x}{0.25} = 4x \qquad (\because h = 0.25)$$

$$y(x) = 1.10100 + 4x\,0.7660625 + \frac{4x(4x+1)}{2!}0.400375 + \frac{4x(4x+1)(4x+2)}{3!}0.09375$$

$$= x^3 + 4.001x^2 + 4.002x + 1.101$$

Put $x = \frac{-1}{3}$

$$y(\frac{-1}{3}) = (\frac{1}{3})^3 + 4.001(\frac{-1}{3})^2 + 4.002(\frac{-1}{3}) + 1.101$$

$$= 0.1745$$

Example 5: Consider the following tabular values:

x	140	150	160	170	180
$y = f(x)$	3685	4845	6302	8076	10225

Determine $y(175)$ using Newton's backward interpolation formula.

Solution:

Let

$$x = 175,\ x_n = 180\ ,h = 10$$

$$r = \frac{x - x_n}{h} = \frac{175 - 180}{10} = -0.5$$

Difference Table:

x	y	∇y_n	$\nabla^2 y_n$	$\nabla^3 y_n$	$\nabla^4 y_n$
140	3685				
		1160			
150	4845		297		
		1457		20	
160	6302		317		38
		1774		58	
170	8076		375		
		2149			
180	10225				

By Newton's backward formula,

$$y_r = y_n + r\nabla y_n + \frac{r(r+1)}{2!}\nabla^2 y_n + \frac{r(r+1)(r+2)}{3!}\nabla^3 y_n + \frac{r(r+1)(r+2)(r+3)}{4!}\nabla^4 y_n + \ldots\ldots\ldots$$

$$y_{175} = 10225 + (-0.5)2145 + \frac{(-0.5)(0.5)375}{2} + \frac{(-0.5)(0.5)(1.5)58}{6} + \frac{(-0.5)(0.5)(1.5)(2.5)38}{24}$$

$$= 9100.5156.$$

INTERPOLATION WITH UNEQUAL INTERVALS:

If the values of x are unequally spaced then interpolation formula for equally spaced points cannot be used. It is, therefore, desirable to develop interpolation formulae for unequally spaced values of x. There are two such formulae for unequally spaced values of x.

(i) Lagrange's interpolation formula

(ii) Newton's interpolation formula with divided difference.

(i) Lagrange's interpolation formula:

Let the function $y = f(x)$ take the values $y_0, y_1, y_2, y_3, ..., y_n$ corresponding to the values $x_0, x_1, x_2, x_3, ..., x_n$ of x. Since there are $(n+1)$ values of x and y, $f(x)$ can be represented by a polynomial in x of degree n.

$$\begin{aligned} y = f(x) = a_0(x-x_1)(x-x_2).....(x-x_n) \\ + a_1(x-x_0)(x-x_2).....(x-x_n) \\ + a_2(x-x_0)(x-x_1).....(x-x_n) \\ + ... \\ + a_n(x-x_0)(x-x_1).....(x-x_{n-1}) \\ .....(1) \end{aligned}$$

Where $a_0, a_1, a_2, ..., a_n$ are constants.

Putting $x = x_0, y = y_0$ in Eq.(1),

$$y_0 = a_0(x_0-x_1)(x_0-x_2).....(x_0-x_n)$$

$$a_0 = \frac{y_0}{(x_0-x_1)(x_0-x_2).....(x_0-x_n)}$$

Similarly,putting $x = x_1, y = y_1$ in Eq.(1)

Proceeding in the same way,

$$a_n = \frac{y_n}{(x_n-x_0)(x_n-x_1).....(x_n-x_{n-1})}$$

Substituting the values of $a_0, a_1, a_2, ..., a_n$

$$f(x) = \frac{(x-x_1)(x-x_2).....(x-x_n)}{(x_0-x_1)(x_0-x_2).....(x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2).....(x-x_n)}{(x_1-x_0)(x_1-x_2).....(x_1-x_n)} y_1 + \\ ..... + \frac{(x-x_0)(x-x_1).....(x-x_{n-1})}{(x_n-x_0)(x_n-x_1).....(x_n-x_{n-1})} y_n$$

This is known as **Lagrange's interpolation formula**.

Example 9: Find the value of y when $x = 10$ from the following table:

x	5	6	9	11
y	12	13	14	16

Solution: By Lagrange's interpolation formula,

$$f(x) = \frac{(x-x_1)(x-x_2).....(x-x_n)}{(x_0-x_1)(x_0-x_2).....(x_0-x_n)} y_0 + \frac{(x-x_0)(x-x_2).....(x-x_n)}{(x_1-x_0)(x_1-x_2).....(x_1-x_n)} y_1 + \\ ..... + \frac{(x-x_0)(x-x_1).....(x-x_{n-1})}{(x_n-x_0)(x_n-x_1).....(x_n-x_{n-1})} y_n$$

$$\begin{aligned} y(10) &= \frac{(10-6)(10-9)(10-11)}{(5-6)(5-9)(5-11)}(12) + \frac{(10-5)(10-9)(10-11)}{(6-5)(6-9)(6-11)}(13) \\ &\quad + \frac{(10-5)(10-6)(10-11)}{(9-5)(9-6)(5-11)}(14) + \frac{(10-5)(10-6)(10-9)}{(11-5)(11-6)(11-9)}(16) \\ &= 2 - 4.3333 + 11.6666 + 5.3333 \\ &= 14.6666 \end{aligned}$$

Example 10: Find the Lagrange interpolating polynomial from the following data:

x	0	1	4	5
$y = f(x)$	1	3	24	39

Solution: By Lagrange's interpolation formula,

$$f(x) = \frac{(x-x_1)(x-x_2).....(x-x_n)}{(x_0-x_1)(x_0-x_2).....(x_0-x_n)}y_0 + \frac{(x-x_0)(x-x_2).....(x-x_n)}{(x_1-x_0)(x_1-x_2).....(x_1-x_n)}y_1 +$$

$$.....+\frac{(x-x_0)(x-x_1).....(x-x_{n-1})}{(x_n-x_0)(x_n-x_1).....(x_n-x_{n-1})}y_n$$

$$f(x) = \frac{(x-1)(x-4)(x-5)}{(0-1)(0-4)(0-5)}(1) + \frac{(x-0)(x-4)(x-5)}{(1-0)(1-4)(1-5)}(3)$$

$$+\frac{(x-0)(x-1)(x-5)}{(4-0)(4-1)(4-5)}(24) + \frac{(x-0)(x-1)(x-4)}{(5-0)(5-1)(5-4)}(39)$$

$$= \frac{1}{20}(3x^3 + 10x^2 + 27x + 20)$$

Example 9: Using Lagrange's formula, express the function $\frac{3x^2+x+1}{(x-1)(x-2)(x-3)}$ as a sum of partial fractions.

Solution: Let us evaluate $y = 3x^2 + x + 1$ for $x = 1, x = 2, x = 3$

x	1	2	3
y	5	15	31

Lagrange's formula is,

$$y = \left(\frac{x-x_1}{x_0-x_1}\right)\left(\frac{x-x_2}{x_0-x_2}\right)y_0 + \left(\frac{x-x_0}{x_1-x_0}\right)\left(\frac{x-x_2}{x_1-x_2}\right)y_1 + \left(\frac{x-x_0}{x_2-x_0}\right)\left(\frac{x-x_1}{x_2-x_1}\right)y_2$$

Substituting the above values, we get

$$y = \left(\frac{x-2}{1-2}\right)\left(\frac{x-3}{1-3}\right)(5) + \left(\frac{x-1}{2-1}\right)\left(\frac{x-3}{2-3}\right)(15) + \left(\frac{x-1}{3-1}\right)\left(\frac{x-2}{3-2}\right)(31)$$

$$y = 2.5(x-2)(x-3) - 15(x-1)(x-3) + 15.5(x-1)(x-2)$$

Thus,

$$\frac{3x^2+x+1}{(x-1)(x-2)(x-3)} = \frac{2.5(x-2)(x-3) - 15(x-1)(x-3) + 15.5(x-1)(x-2)}{(x-1)(x-2)(x-3)}$$

$$= \frac{2.5}{x-1} - \frac{15}{x-2} + \frac{15.5}{x-3}$$

Example: Given the values

x:	5	7	11	13	17
f(x):	150	392	1452	2366	5202

Evaluate f(9), using Lagrange's formula.

Example: A curve passes through the points (0,18), (1,10), (3,-18) and (6, 90). Find the slope of the curve at x=2.

DIVIDED DIFFERENCES:

In Lagrange's interpolation formula, if another interpolation value is added then the interpolation

coefficients are required to be recalculated. To avoid this recalculation, Newton's general interpolation formula is used.

If $(x_0, y_0), (x_1, y_1), ..., (x_n, y_n)$ be given points then the first divided difference for x_0, x_1 is defined by the relation,

$$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0}$$

Similarly, $[x_1, x_2] = \dfrac{y_2 - y_1}{x_2 - x_1}$, etc.

The second divided difference for x_0, x_1, x_2 is defined as

$$[x_0, x_1, x_2] = \frac{[x_1, x_2] - [x_0, x_1]}{x_2 - x_0}$$

The third divided difference for x_0, x_1, x_2, x_3 is defined as

$$[x_0, x_1, x_2, x_3] = \frac{[x_1, x_2, x_3] - [x_0, x_1, x_2]}{x_3 - x_0}$$ and so on.

NEWTON'S DIVIDED DIFFERENCE FORMULA:

Let the function $y = f(x)$ take the values $y_0, y_1, y_2, y_3, ..., y_n$ corresponding to the values $x_0, x_1, x_2, x_3, ..., x_n$ respectively. According to the definition of divided differences,

$$[x_0, x_1] = \frac{y_1 - y_0}{x_1 - x_0}$$

$$y = y_0 + (x - x_0)[x, x_1] \text{.....(1)}$$

$$[x, x_0, x_1] = \frac{[x, x_0] - [x_0, x_1]}{x - x_1}$$

$$[x, x_0] = [x_0, x_1] + (x - x_1)[x, x_0, x_1]$$

Substituting in Eq.(1),

$$y = y_0 + (x - x_0)\{[x_0, x_1] + (x - x_1)[x, x_0, x_1]\}$$

$$= y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x, x_0, x_1]$$

$$\text{.....(2)}$$

Also, $[x, x_0, x_1, x_2] = \dfrac{[x, x_0, x_1] - [x_0, x_1, x_2]}{x - x_2}$

$$[x, x_0, x_1] = [x_0, x_1, x_2] + (x - x_2)[x, x_0, x_1, x_2]$$

Substituting in Eq.(2)

$$y = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x, x_0, x_1]$$

$$= y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)\{[x_0, x_1, x_2] + (x - x_2)[x, x_0, x_1, x_2]\}$$

$$= y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2]$$

$$+ (x - x_0)(x - x_1)(x - x_2)[x, x_0, x_1, x_2]$$

and so on.

Finally, we have

$$y = y_0 + (x - x_0)[x_0, x_1] + (x - x_0)(x - x_1)[x_0, x_1, x_2]$$

$$+ (x - x_0)(x - x_1)(x - x_2)[x_0, x_1, x_2, x_3] + \ldots\ldots$$

$$+ (x - x_0)(x - x_1)\ldots\ldots(x - x_{n-1})[x, x_0, x_1, x_2, \ldots, x_n] \text{.....(3)}$$

Equation (3) is known as **Newton's general interpolation formula** for divided differences.

Example 11: Using Newton's divided difference interpolation, compute the value of $f(6)$ from the table given below:

x	1	2	7	8
$y = f(x)$	1	5	5	4

Solution:

Divided Difference Table:

x	$f(x)$	1st DD	2nd DD	3rd DD
1	1			
		4		
2	5		$-\frac{2}{3}$	
		0		$\frac{1}{14}$
7	5		$-\frac{1}{6}$	
		-1		

8	4			

By Newton's divided difference formula,

$$y = y_0 + (x-x_0)[x_0,x_1] + (x-x_0)(x-x_1)[x_0,x_1,x_2] + (x-x_0)(x-x_1)(x-x_2)[x_0,x_1,x_2,x_3] + \ldots\ldots$$
$$+ (x-x_0)(x-x_1)\ldots\ldots(x-x_{n-1})[x,x_0,x_1,x_2,\ldots,x_n]$$

$$f(6) = 1 + (6-1)(4) + (6-1)(6-2)\left(-\frac{2}{3}\right) + (6-1)(6-2)(6-7)\left(\frac{1}{14}\right)$$
$$= 6.2381$$

Example 12: Using Newton's divided difference interpolation formula, find a polynomial.

x	1	2	4	7
$y = f(x)$	10	15	67	430

Solution:

Divided Difference Table:

x	$f(x)$	1st DD	2nd DD	3rd DD
1	10			
		5		
2	15		7	
		26		2
4	67		19	
		121		
7	430			

By Newton's divided difference formula,

$$y = y_0 + (x-x_0)[x_0,x_1] + (x-x_0)(x-x_1)[x_0,x_1,x_2] + (x-x_0)(x-x_1)(x-x_2)[x_0,x_1,x_2,x_3] + \ldots\ldots$$
$$+ (x-x_0)(x-x_1)\ldots\ldots(x-x_{n-1})[x,x_0,x_1,x_2,\ldots,x_n]$$

$$f(x) = 10 + (x-1)(5) + (x-1)(x-2)(7) + (x-1)(x-2)(x-4)(2)$$
$$= 2x^3 - 7x^2 + 12x + 3$$

Example:

Given the values

x:	5	7	11	13	17
f(x):	150	392	1452	2366	5202

Evaluate f(9), using Newton's divided difference formula.

Example: Using Newton's divided difference formula, find the missing value from the table:

x:	1	2	4	5	6
y:	14	15	5	?	9

Parul® University
Vadodara, Gujarat

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